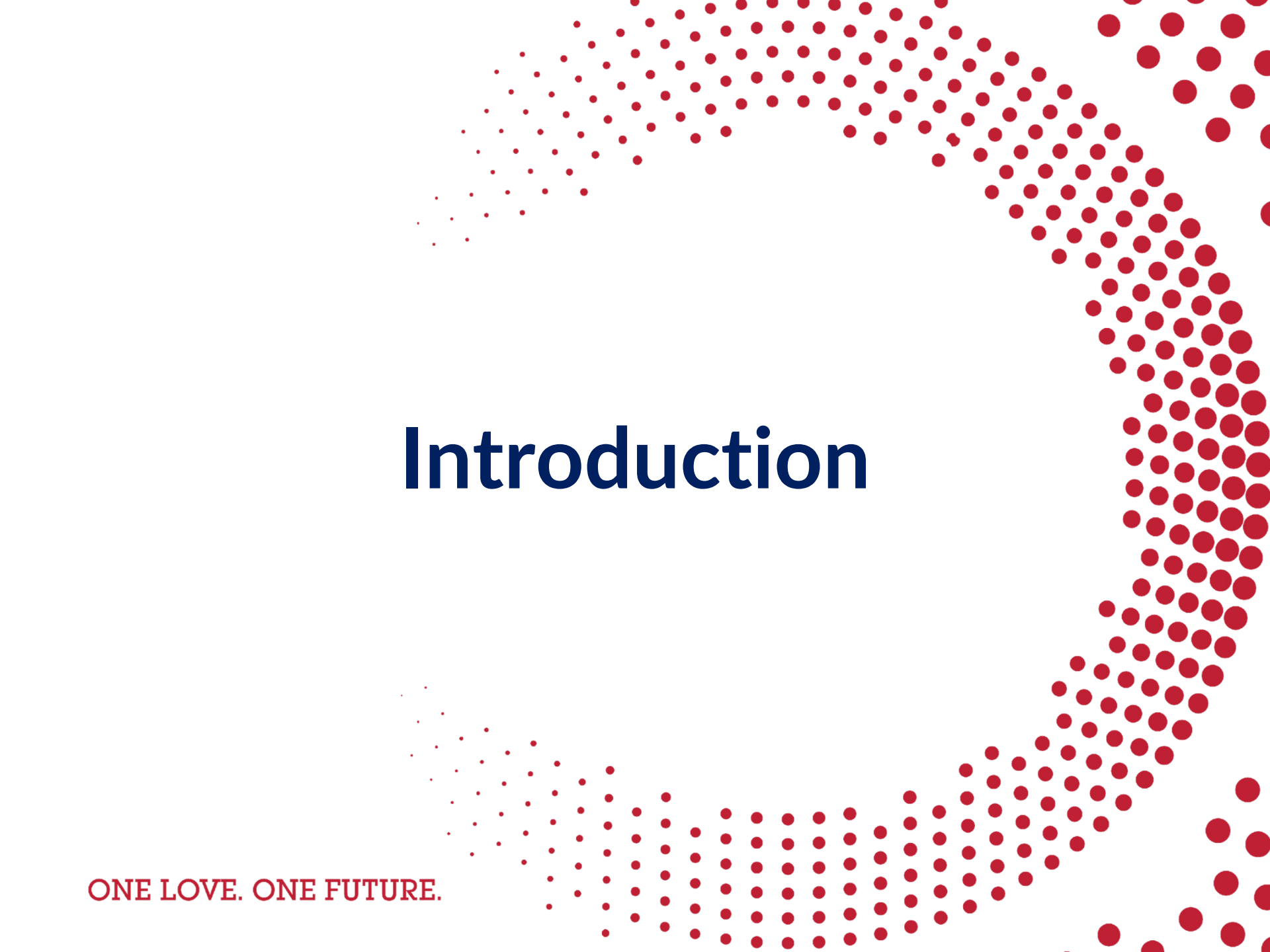


A decorative background consisting of a large number of red dots of varying sizes. These dots are arranged to form a circular shape that is open on the left side, with the dots being more densely packed on the right side.

Introduction

ONE LOVE. ONE FUTURE.

Contents

1. Introduction and Career
2. Internet and Web
3. Browser and How the Web Works
4. Protocols and Standards
5. Hosting and Deployment

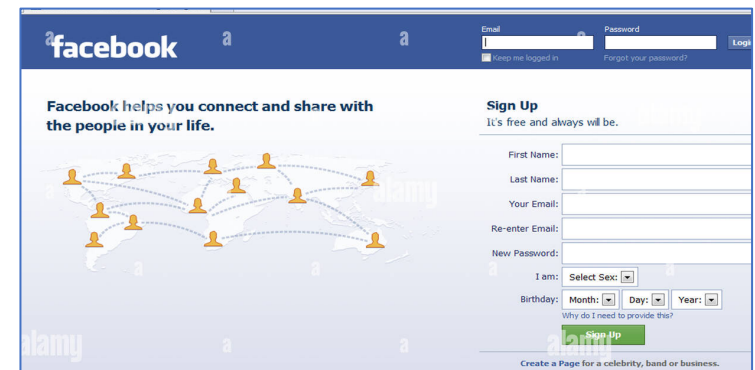
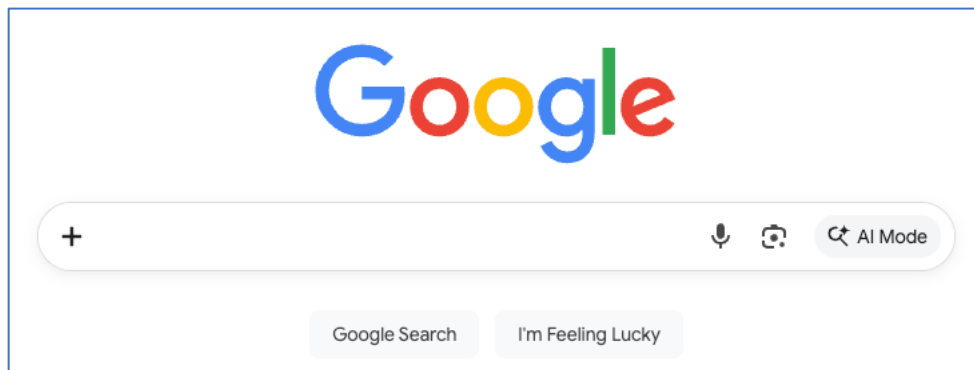


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1. Introduction and Career

Why Web matters today

- Web is everywhere
 - Billions of users access web services every day
 - Most digital products run on the Web
 - Runs across devices: desktop, mobile, embedded
- Web powers the digital economy
 - E-commerce, banking, education, social media
 - Businesses rely on web presence



Web Development

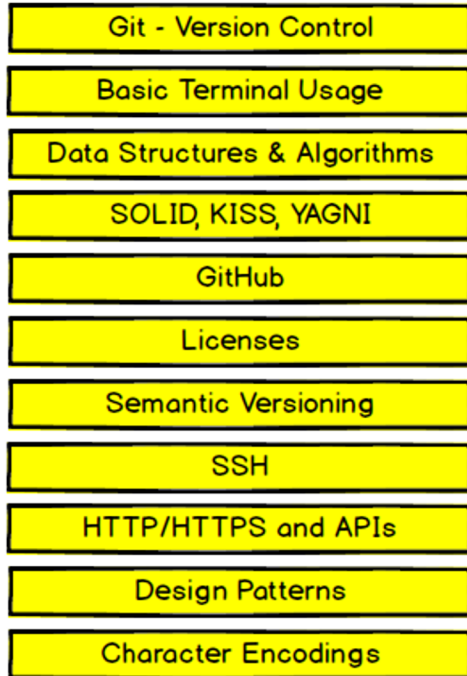
- Key areas of Web development
 - Frontend – user interface and experience
 - Backend – business logic and data processing
 - Database – storing and managing data
 - Infrastructure – hosting, deployment, scalability
- Modern trends
 - Single-page applications (SPA)
 - Real-time web apps
 - Microservices and serverless architectures
 - AI-powered features

Web Developer Roles

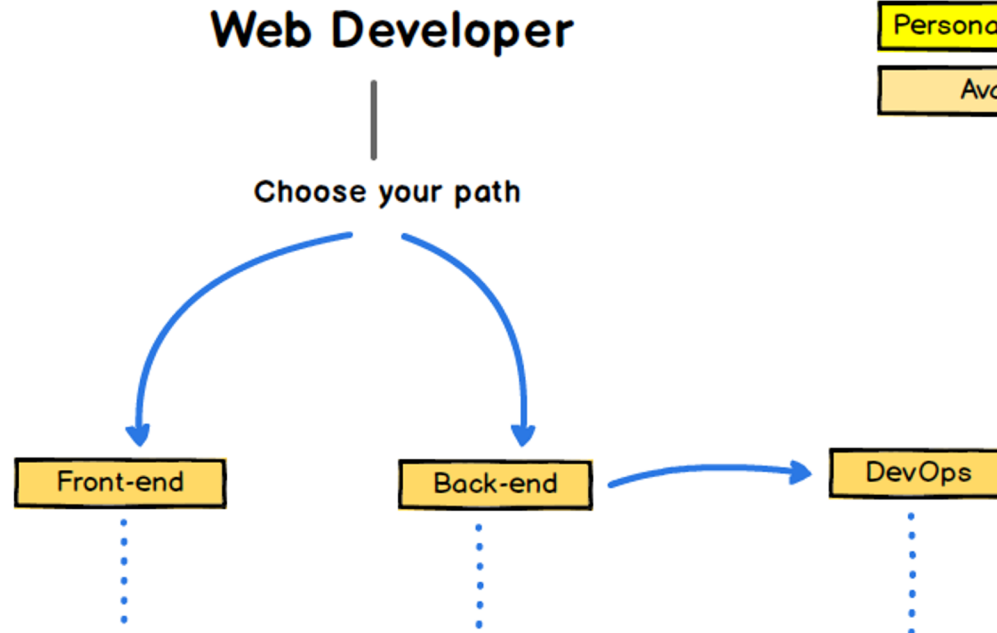
- Frontend Developers: focus on the user interface and experience
 - Build layouts and interactions
 - Work with browsers and APIs
- Backend Developers: focus on server-side logic and data
 - Implement business logic
 - Design APIs
 - Manage databases
- Full-stack Developers: combine frontend and backend skills
- DevOps Engineer: focus on running and operating systems

Roadmap Overview

Required for any path



Legends



<https://roadmap.sh/>

Frontend Path



Related Roadmaps

- ✓ JavaScript Roadmap
- ✓ React Roadmap
- ✓ TypeScript Roadmap
- ✓ Node.js Roadmap

HTML, CSS and JavaScript are the backbone of web development. Make sure to practice by building lots of projects.

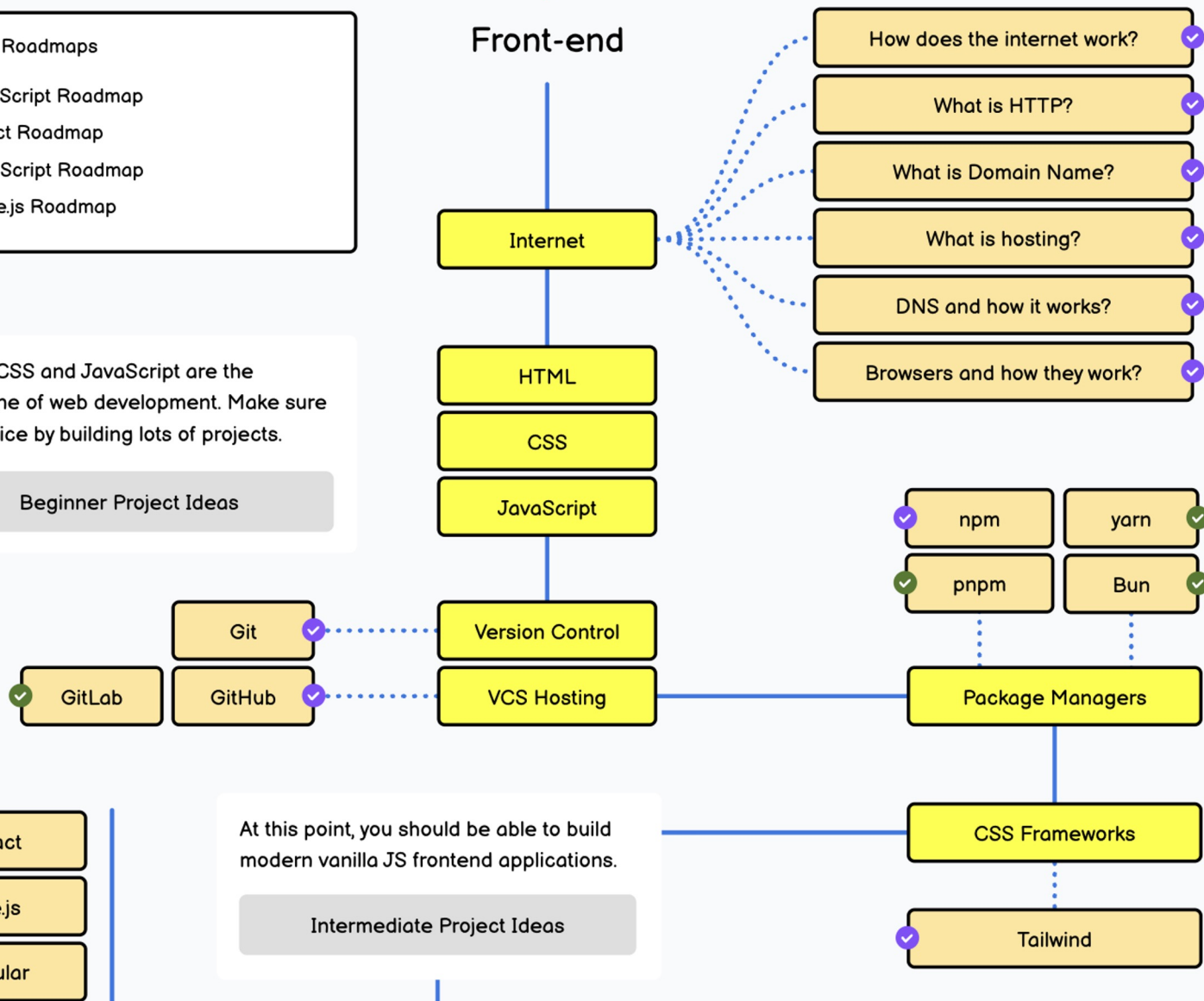
Beginner Project Ideas

- ✓ React
- ✓ Vue.js
- ✓ Angular

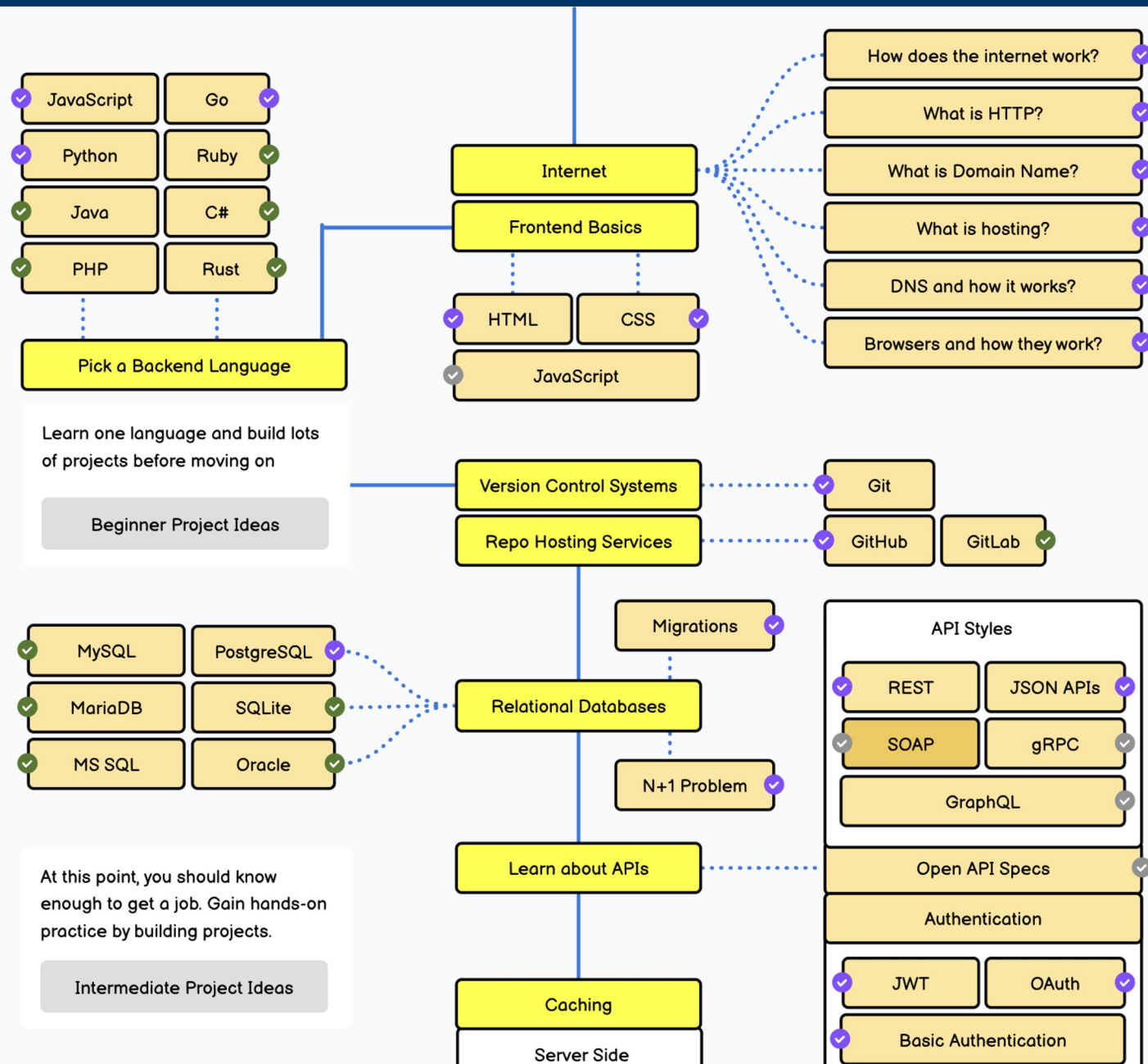
At this point, you should be able to build modern vanilla JS frontend applications.

Intermediate Project Ideas

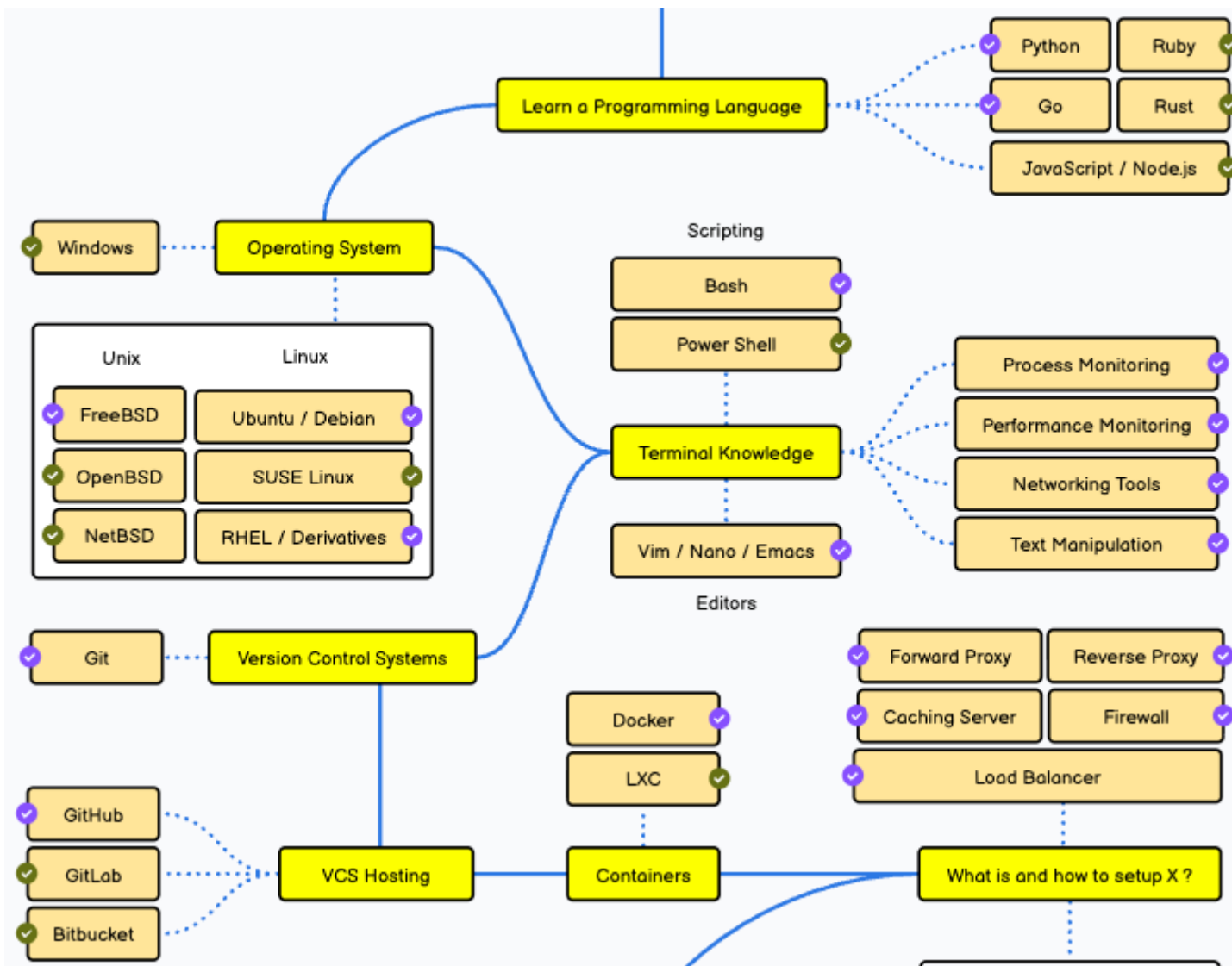
Front-end



Backend Path



DevOps Path



Foundation Skills for Web Development

- Working with Code and Projects
 - Using **Git** to track changes
 - Cloning, committing, and pushing code
- Understanding How the Web Communicates
 - Reading HTTP requests and responses
 - Using browser **DevTools** to inspect network activity
- Running Applications Locally
 - Using **Terminal** to navigate and execute commands
 - Starting development servers
- Preparing Software to go live
 - Setting environment using **Docker**
 - Deploying applications on **Web hosting services**

Job Opportunity (2025)

Criteria	Web Developer	AI / Machine Learning	Data Science
Job opportunities	Very high demand	Growing demand	Strong demand in fintech, commerce
Fresher (0–1 yr)	~9–15M VND/month	~20–35M VND/month	~15–30M VND/month
Junior (1–3 yrs)	~15–30M	~30–60M	~25–45M
Senior (5+ yrs)	~35–70M+	~60–120M+	~45–80M+
Ease of learning	Easiest entry, fast product	Hot job (but fewer fresher positions)	Strong business relevance
Technical difficulty	Many technologies	Hard (Math, ML theory)	Statistics + Coding

Web Developer in AI era

- AI will replace repetitive coding – not developers who understand systems.
- AI accelerates development, but does not replace engineering.
 - Web systems are complex (architecture, integration, scalability)
 - Businesses require custom logic and domain understanding
 - Production software needs reliability, security, and maintenance
- What Web Developers should do
 - Shift from coding only → building systems.
 - Focus on problem solving, not syntax
 - Use AI as a productivity tool (copilot, prototyping, debugging)



2. Internet and Web

Questions:

- What is the Internet?
- What is the World Wide Web?

Web $\neq$ Internet

- The Internet is a global physical network that connects computers and devices using standardized protocols (such as TCP/IP) to share and transmit information.
- The World Wide Web is a system of interlinked hypermedia documents and resources stored on servers connected to the Internet and accessed through web browsers using protocols such as HTTP/HTTPS



History of the Internet

- The idea of a long-distance computer network traces back to early 60's
 - Joseph Licklider at M.I.T. (a “time-sharing network of computers”)
 - Paul Baran at Rand (tasked with designing a “survivable” communications system from a nuclear attack)
 - Donald Davies at National Physics Laboratory in U.K.
- US Department of Defense was interested in the development of distributed, decentralized networks
 - survivability (i.e., network still functions despite a local attack)
 - fault-tolerance (i.e., network still functions despite local failure)
 - contrast with phone system, electrical system which are highly centralized services.

The Internet

- In 1969, Advanced Research Project Agency funded the ARPANET
 - Connected computers at UC Los Angeles, UC Santa Barbara, Stanford Research Institute, and University of Utah
 - Allowed researchers to share data, communicate
- Technical origin
 - One of earliest attempts to network heterogeneous, geographically dispersed computers
 - Email first available on ARPANET in 1971 (and quickly very popular!)

Vietnam Internet History



Stage	Timeline	Characteristics
Trial and Initial Phase	Before 1997	IOIT - AUS (1994), VDC - Sprintlink (1996)
Official Launch	19/11/1997	VNPT, Netnam, FPT..
Growth	Present	80% of the population

History of the Web

- In 1989, Tim Berners-Lee designed a hypertext system for linking documents over the internet
 - Designed a (non-wysiwyg) language for specifying document content => Evolved into hypertext markup language (HTML)
 - Designed a protocol for downloading documents and interpreting the content => Evolved into hypertext transfer protocol (HTTP)
- Today, the Web is the most visible aspect of the Internet

IP Address & Port

- Idea: Internet connects devices (IP) and services (Port).
- IP Address
 - A unique identifier for a device on a network
 - Used to locate servers and clients on the Internet
 - Examples: 8.8.8.8
- Port
 - A number that identifies a specific service on a device
 - Allows multiple applications to run on the same IP
 - Common examples:
 - 80 → HTTP
 - 443 → HTTPS
 - 22 → SSH

Internet Services

- Common Internet Services
 - Email — sending and receiving messages
 - File Transfer — sharing
 - Remote Access — controlling computers remotely
 - Messaging & VoIP — chat and video calls
- Internet = infrastructure
- Web = one application running on it

Directions of Web

- Web 1.0 — Read-Only Web (1990s – early 2000s)
 - Static pages with limited interaction
 - Examples: personal websites, early company pages
 - Users consume content
- Web 2.0 — Interactive & Social Web (2000s – present)
 - Dynamic websites and user-generated content
 - Examples: Facebook, YouTube, e-commerce,
 - Users create and interact
- Vision
 - Semantic Web (Tim Berners-Lee's vision)
 - Web3 (Blockchain community)



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3. Browser & How the Web work?

Web Browser

- The Web was an obscure, European research tool until 1993
- In 1993, Marc Andreessen and Eric developed Mosaic, one of the early graphical Web browsers
- Andreessen left NCSA to found Netscape in 1994
- 1995, Microsoft came out with Internet Explorer
- 1996, Opera web browser
- 2004, Firefox web browser, version 1.0
- 2008, Google Chrome
- Today, the Web is the most visible aspect of the Internet

Web Browsers - History



IE crew in front of Netscape building
IE 4.0, 1997



The next day

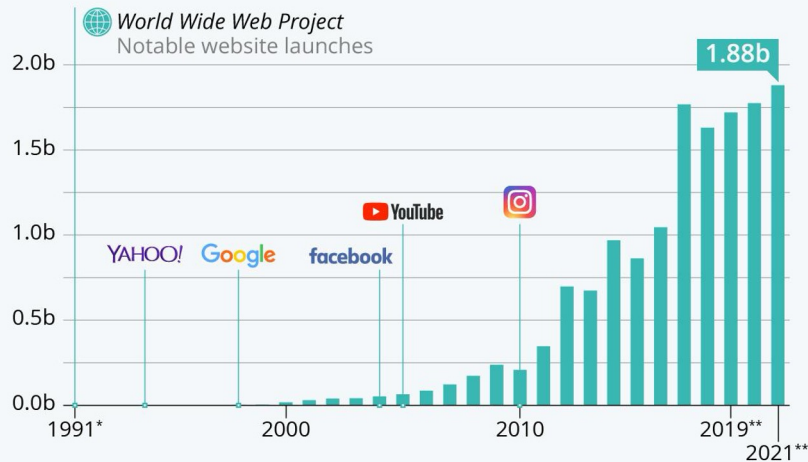
Browser Responsibilities

- A web browser acts as the interface between users and the Web
 - Request resources
 - Convert URLs into HTTP/HTTPS requests
 - Communicate with servers
 - Send requests and receive responses
 - Render content
 - Interpret HTML, CSS, and JavaScript to display pages
 - Manage user interaction
 - Handle input, navigation, forms, and events

Web Growth

How Many Websites Are There?

Number of websites online from 1991 to 2021



* As of August 1, 1991.

** Latest available data for 2019: October 28, for 2020: June 2, for 2021: August 6.

Source: Internet Live Stats



statista

Google Chrome →

Firefox →

Safari →

IE, Opera →

Netscape →

Mosaic →

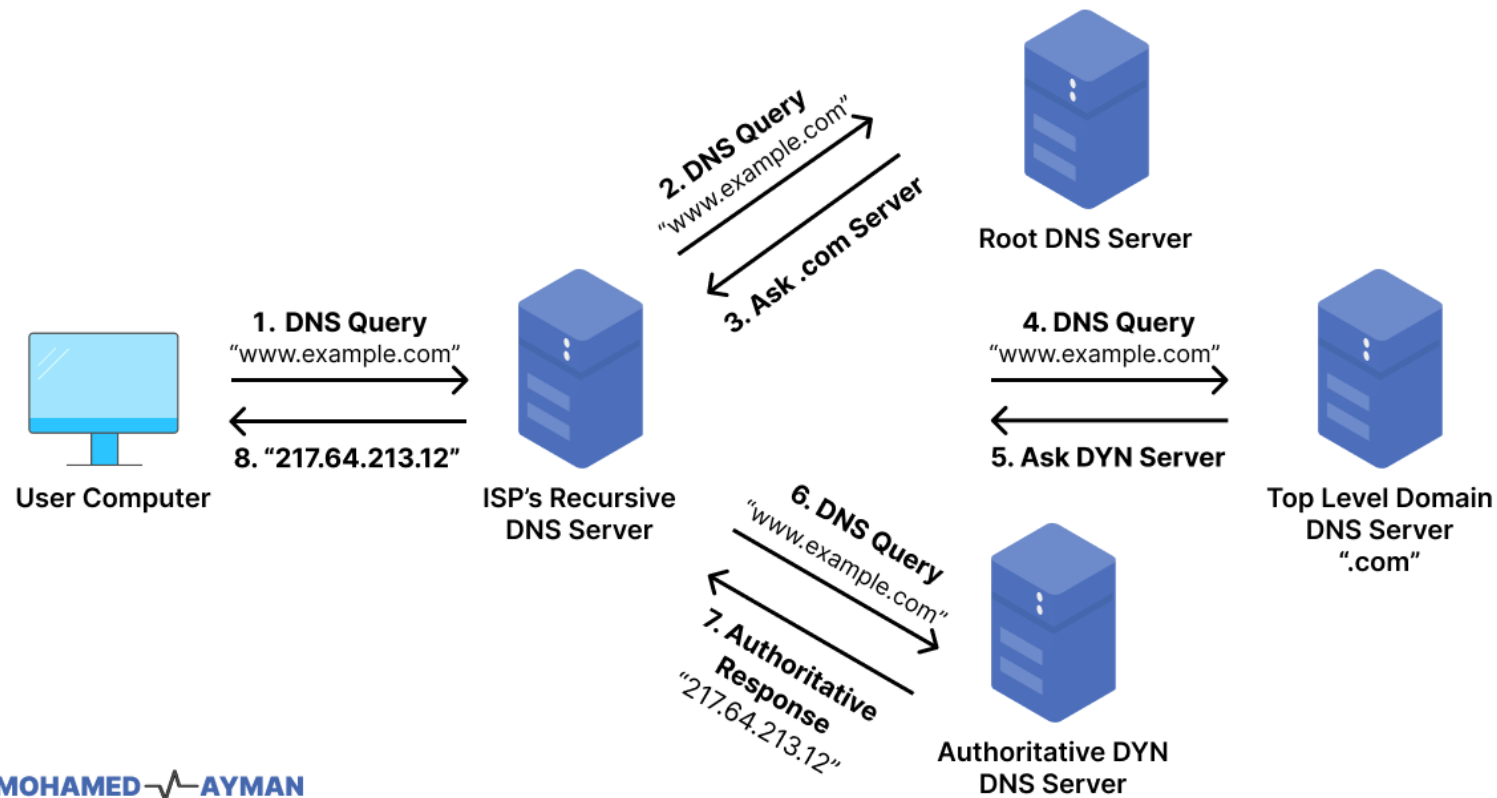
Year	Computers on the Internet	Web Servers on the Internet
2011	~605,000,000	~250,000,000
2008		172,338,726
2006	439,286,364	85,541,228
2004	285,139,107	56,923,737
2002	162,128,493	33,082,657
2000	93,047,785	18,169,498
1998	36,739,000	4,279,000
1996	12,881,000	300,000
1994	3,212,000	3,000
1992	992,000	50

What Happens When You Type a URL?

- DNS lookup — find the server's IP address
- Connection established — browser connects to the server
- Request sent — browser asks for resources
- Server processing — server prepares a response
- Response returned — data is sent back
- Rendering — browser displays the page

Step 1. DNS Lookup

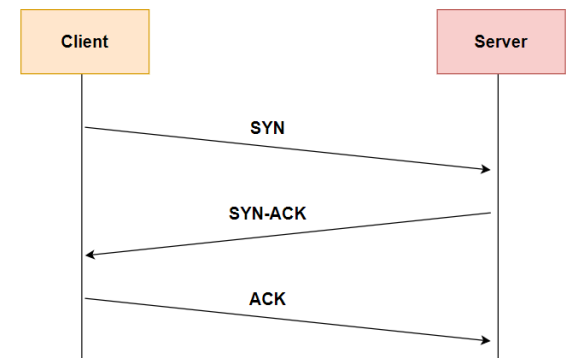
- DNS: Domain Name System



MOHAMED AYMAN

Step 2 – TCP Connection

- The browser opens a TCP connection to the server's IP and port
 - This ensures reliable communication
 - Both sides agree on how data will be exchanged
- Three-Way Handshake (Simplified)
 - Client → SYN (request to connect)
 - Server → SYN-ACK (acknowledges)
 - Client → ACK (connection established)



Step 3: HTTP Request (High Level)

- Browser sends a request
 - Browser asks the server for a resource.
 - The request includes the page URL
 - It specifies an action (e.g., retrieve data, submit data)
- What the server receives in a request
 - What resource is needed
 - How it should be processed
 - Who is making the request (cookies, authentication)
- Examples of requests
 - Loading a webpage
 - Submitting a form

Step 4 – Server Processing

- What the Server does
 - Validates the request
 - Executes application logic
 - Retrieves or updates data (database, APIs)
 - Generates content (HTML, JSON, files)
- Types of responses
 - Static response – return pre-built files
 - Dynamic response – generate content based on user, data, or logic
- Why it matters
 - Determines performance and scalability
 - Enables personalization and business logic

Step 5 – HTTP Response (High Level)

- What the response contains
 - Status code – result of the request (e.g., 200, 404)
 - Headers – metadata about the response
 - Body – actual content (HTML, JSON, images, files)
- What happens next
 - The browser receives the response
 - Additional resources may be requested (CSS, JS, images)
 - The rendering process begins
- Why it matters
 - Defines whether the request succeeded
 - Provides data for the user interface
 - Impacts performance and user experience

Step 6 – Browser Rendering

- What happens
 - Parse HTML → build the DOM
 - Load CSS → compute layout and styles
 - Execute JavaScript → update content and behavior
 - Render pixels on the screen
- Rendering is incremental
 - Pages may appear gradually
 - Additional resources load asynchronously
- Why it matters
 - Determines user experience and performance
 - Affects perceived speed of a website
 - Influences design decisions



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4. Web Protocols & Standards

DNS - Record Types

- Common DNS Record Types
 - A Record — maps a domain to an IPv4 address
 - AAAA Record — maps a domain to an IPv6 address
 - CNAME — maps one domain to another domain
 - MX Record — defines mail servers for a domain
 - TXT Record — stores verification or configuration data
- Examples
 - hust.edu.vn → A → 202.191.59.134
 - dobalam.id.vn → TXT → hosting-site=hust-cv-student

Domain Name Structure

- A domain name is a human-readable address that identifies a website
- soict.hust.edu.vn
 - subdomain: soict
 - second-level domain: hust
 - domain category: edu
 - country-top level domain: vn

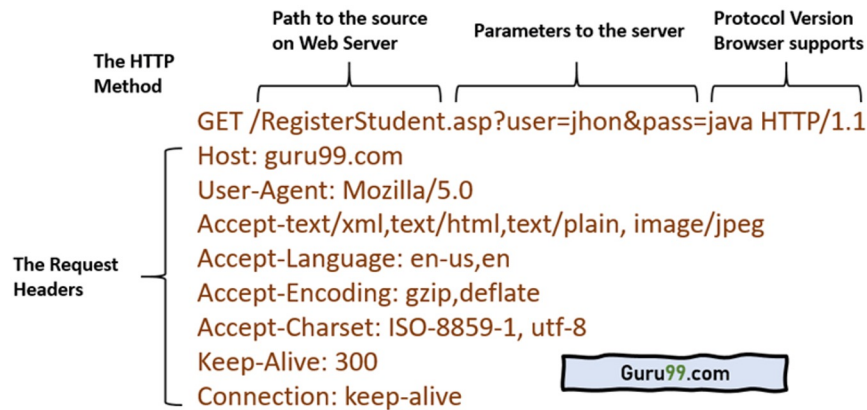
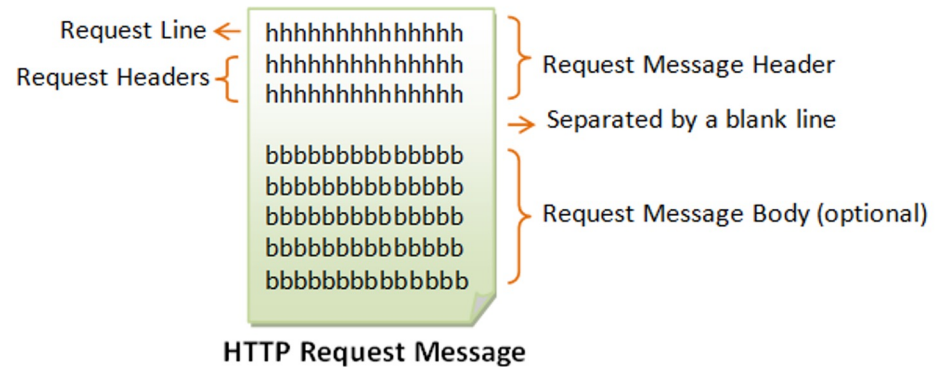
- **URI (Uniform Resource Identifier)** is the general concept for identifying resources on the Internet
- **Uniform Resource Name (URN)**
 - Identifies a resource by a unique name
 - Often used in standards and libraries
 - Example: urn:isbn:0451450523
- **Uniform Resource Locator (URL)**
 - Identifies where a resource is located
 - Includes protocol, domain, and path
 - Example: <https://example.com/page>

Hypertext Transport Protocol (HTTP)

- HTTP is based on the request-response communication model:
 - Client sends a request
 - Server sends a response
- HTTP is a stateless protocol
 - The protocol does not require the server to remember anything about the client between requests.

HTTP Request

- Structure of the request:
 - request line
 - request headers
 - blank line
 - optional request body



HTTP Request

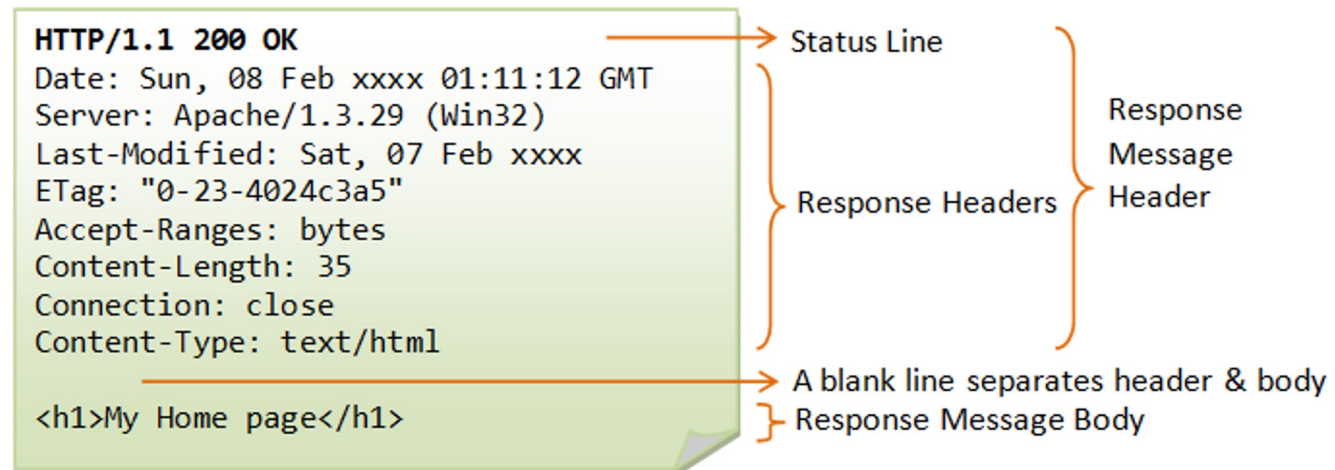
- Request line
 - GET /test.html HTTP/1.1
 - POST /index.html HTTP/1.1
- Three space-separated parts:
 - HTTP request method
 - Request-URI (Uniform Resource Identifier)
 - HTTP version

HTTP Request

- Request line
 - GET /test.html HTTP/1.1
 - POST /index.html HTTP/1.1
- Three space-separated parts:
 - HTTP request method
 - Request-URI
 - **HTTP version**
 - 1.1: 1997
 - 2: 2015
 - 3: 2022

HTTP Response

- Structure of the response:
 - status line
 - header field(s)
 - blank line
 - optional body



HTTP Response

- Status line
 - Example: HTTP/1.1 200 OK
- Three space-separated parts:
 - HTTP version
 - status code
 - reason phrase (intended for human use)

HTTP Response

- Status code
 - Three-digit number
 - First digit is class of the status code:
 - 1=Informational
 - 2=Success
 - 3=Redirection (alternate URL is supplied)
 - 4=Client Error
 - 5=Server Error
 - Other two digits provide additional information
 - See <http://www.w3.org/Protocols/rfc2616/rfc2616-sec10.html>

HTTP Response

- Common header fields:
 - Connection, Content-Type, Content-Length
 - Date: date and time at which response was generated (required)
 - Location: alternate URI if status is redirection
 - Last-Modified: date and time the requested resource was last modified on the server
 - Expires: date and time after which the client's copy of the resource will be out-of-date
 - ETag: a unique identifier for this version of the requested resource (changes if resource changes)

HTTP Request/Response Examples

Connect	{	\$ telnet www.example.org 80 Trying 192.0.34.166... Connected to www.example.com (192.0.34.166). Escape character is '^['.
Send Request	{	GET / HTTP/1.1 Host: www.example.org
Receive Response	{	HTTP/1.1 200 OK Date: Tue, 11 Oct 2022 20:30:49 GMT ...

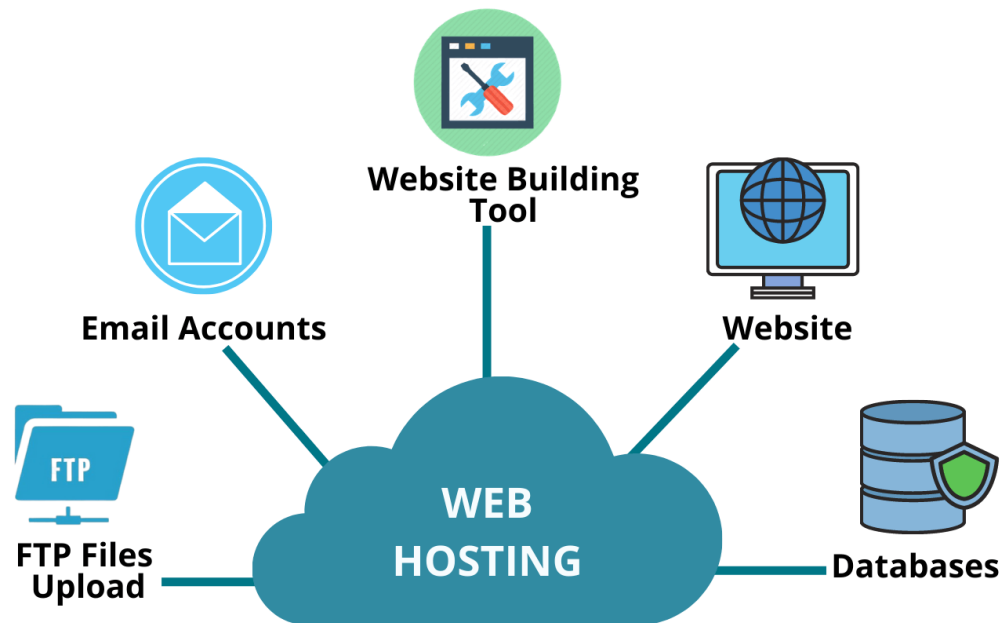


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4. Web Hosting & Deployment

Web Hosting

- Web hosting is a service that provides infrastructure to run websites
- It may be cloud-based or traditional

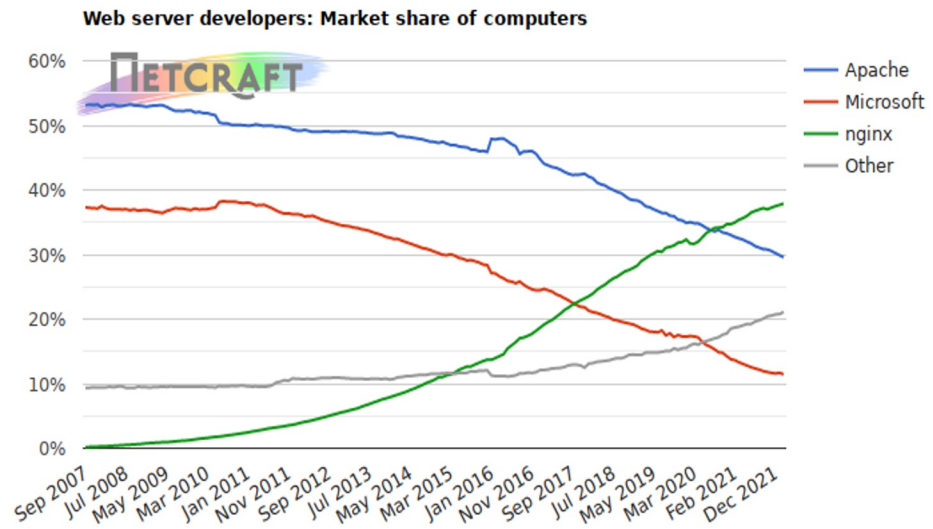


Web Hosting: main types

1. Shared hosting: multiple websites are stored on a same server
2. Virtual Private Server (VPS) hosting: a physical server is split into many VPS servers, each with its dedicated resources and OS.
3. Dedicated hosting: rents an entire physical server for your website
4. Cloud hosting: utilizes a network of interconnected servers to host a website.
5. Free web hosting: free service

Web Servers (Software)

- Software responsible for handling HTTP/HTTPS requests from browsers and sending responses (usually web pages) back to users.
- Main functions:
 - Receive requests from clients (browsers).
 - Process web files (HTML, CSS, JS, PHP...).
 - Return responses (web pages, data) to clients.
- Examples:
 - Apache HTTP Server
 - Nginx
 - Microsoft IIS



Web Server vs Hosting

- Web Server (software)
 - Software that handles HTTP/HTTPS requests
 - Run on hardware that stores web application
 - Processes web files and returns responses
 - Example: Apache, Nginx, IIS
 - Focus: serving web content
- Web Hosting (service)
 - Service that provides infrastructure to run a website
 - Includes servers, storage, networking, and management
 - Offered by providers like AWS, Google Cloud...
 - Focus: where the website lives



Exercises

- Register a free domain name
 - <https://www.pavietnam.vn/vn/tin-tuc-uu-dai-so-huu-mien-phi-ten-mien-id-vn.html>
 - <https://www.matbao.net/ten-mien/ten-mien-mien-phi.html>
- Git - GitHub
 - <https://codelearn.io/sharing/git-github-tu-co-ban-den-nang-cao-p1>
 - <https://docs.github.com/en/get-started/start-your-journey/about-github-and-git>

